



User Guide & Running Instructions

Air Quality Prediction Project — step-by-step for first-time users

This guide assumes you have **never** run a machine-learning project before. Follow the steps in order. The easiest path is **Anaconda + Jupyter Notebook**.

Part A — Install the software (one time only)

Step 1: Install Anaconda (includes Python + Jupyter).

Go to <https://www.anaconda.com/download>, download the version for your operating system, and install it like any normal program (keep clicking *Next*). Anaconda installs Python and Jupyter Notebook together, so you do not need to install Python separately.

Step 2: Open the Anaconda Prompt.

On Windows, click Start and search for Anaconda Prompt. On macOS/Linux open the normal Terminal. This black window is where you type commands.

Step 3: Install the required libraries.

Type this single command and press Enter (it downloads pandas, scikit-learn, etc.):

```
pip install -r requirements.txt
```

Tip: first move into the Source_Code folder, e.g. `cd Air_Quality_Prediction_Project/Source_Code`, then run the command above. If pip gives an error, try pip3 instead.

Part B — Open and run the project

Step 4: Launch Jupyter Notebook.

In the same prompt type:

```
jupyter notebook
```

Your web browser opens automatically showing your files.

Step 5: Open the notebook.

Click into `Source_Code` and open `Air_Quality_Prediction.ipynb`.

Step 6: Run the code.

From the top menu choose **Cell → Run All** (or click each cell and press **Shift + Enter** one by one). The notebook loads the dataset, cleans it, trains the models and shows the graphs automatically.

Step 7: Loading the dataset.

You do not need to do anything special — the notebook already reads the file from `../Dataset/AirQualityUCI.csv`. Just keep the folder structure as provided.

Part C — Understanding the output

- **Tables** show the data and the model scores.
- **R² score** close to 1.0 means the model is accurate. In this project the Random Forest reaches about **0.90**.
- **MAE / RMSE** are the average error sizes — smaller is better.
- **Graphs** show daily pollution trends, correlations, model comparison and a predicted-vs-actual plot.
- The final cell prints a sample **prediction** next to the actual value so you can see the model working.

Part D — Fixing common errors

Error message	What it means & how to fix
<code>ModuleNotFoundError: No module named 'pandas'</code>	A library is missing. Re-run <code>pip install -r requirements.txt</code> .
<code>FileNotFoundError: AirQualityUCI.csv</code>	The dataset moved. Keep the <code>Dataset</code> folder next to <code>Source_Code</code> exactly as provided.
Kernel keeps "busy" / stuck	Click Kernel → Restart & Run All .
<code>pip is not recognized</code>	Use the Anaconda Prompt (not the normal Command Prompt), or try <code>pip3</code> .

Alternative: running the plain script

If you prefer not to use the notebook, open `air_quality_prediction.py` in VS Code, or run it from the prompt inside the Source_Code folder:

```
python air_quality_prediction.py
```

It prints the model comparison and saves the graphs into the `figures` folder.